EXAM PAPERS PRACTICE

AQA A Level Further Mathematics (7367)

Topic **C: Matrices**

Sub topic **C6: Singular Matrices and Inverse of a 2x2 Matrix**

Topic Questions

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Question Paper

Topic **C: Matrices**

Sub topic **C6: Singular Matrices and Inverse of a 2x2 Matrix**

To be used by all students and teachers for AQA A Level Further Mathematics (7367) for exams from 2025 onwards.

Students and teachers of other boards may also find this useful

Name : ______________________

Date : ______________________

Time : ______________________

Total Marks Available : _____

Total Marks Received : _____

EXAM PAPERS PRACTICE

Q1.

Two matrices **A** and **B** satisfy the equation

$$\mathbf{AB} = I + 2\mathbf{A}$$

where I is the identity matrix and $\mathbf{B} = \begin{bmatrix} 3 & -2 \\ -4 & 8 \end{bmatrix}$

Find **A**.

(Total 3 marks)

Q2.

$\mathbf{A} = \begin{bmatrix} 1 & 2 \\ 1 & k \end{bmatrix}$ $\quad \mathbf{B} = \begin{bmatrix} -1 & 0 \\ 0 & 1 \end{bmatrix}$

(a) Find the value of k for which matrix **A** is singular.

(1)

(b) Describe the transformation represented by matrix **B**.

(1)

(c) (i) Given that **A** and **B** are both non-singular, verify that $\mathbf{A}^{-1}\mathbf{B}^{-1} = (\mathbf{BA})^{-1}$.

(4)

(ii) Prove the result $\mathbf{M}^{-1}\mathbf{N}^{-1} = (\mathbf{NM})^{-1}$ for all non-singular matrices **M** and **N**.

(4)

(Total 10 marks)

Q3.

The transformation T is defined by the matrix **M**. The transformation S is defined by the matrix $\mathbf{M}^{-1}$. Given that the point (x, y) is invariant under transformation T, prove that (x, y) is also an invariant point under transformation S.

(Total 3 marks)

Q4.

A student claims:
"Given any two non-zero square matrices, **A** and **B**, then $(\mathbf{AB})^{-1} = \mathbf{B}^{-1}\mathbf{A}^{-1}$ "

(a) Explain why the student's claim is incorrect giving a counter example.

(2)

(b) Refine the student's claim to make it fully correct.

(1)

(c) Prove that your answer to part **(b)** is correct.

(3)

(Total 6 marks)

Q5.

The plane transformation T is a rotation through θ radians anticlockwise about O, and maps points (x, y) onto image points (X, Y) such that

$$\begin{bmatrix} X \\ Y \end{bmatrix} = \begin{bmatrix} c & -s \\ s & c \end{bmatrix} \begin{bmatrix} x \\ y \end{bmatrix}$$

where $c = \cos\theta$ and $s = \sin\theta$.

(a) Write down the inverse of the matrix $\begin{bmatrix} c & -s \\ s & c \end{bmatrix}$ and hence show that

$$x = cX + sY \quad \text{and} \quad y = -sX + cY$$

(3)

(b) The curve C has equation $x^2 - 6xy - 7y^2 = 8$.

The image of C under T is the curve C' with equation $pX^2 + qXY + rY^2 = 8$.

(i) Use the results of part (a) to show that

$$q = 6s^2 + 16sc - 6c^2$$

and express p and r similarly in terms of c and s.

(4)

(ii) Given that θ is an acute angle, find the values of c and s for which $q = 0$ and hence in this case express the equation of C' in the form

$$\frac{X^2}{a^2} - \frac{Y^2}{b^2} = 1$$

(8)

(iii) Hence explain why C is a hyperbola.

(1)

(Total 16 marks)

Q6.

(a) (i) Find the eigenvalues and corresponding eigenvectors of $\mathbf{A} = \begin{bmatrix} 1 & 3 \\ -2 & 8 \end{bmatrix}$.

(6)

(ii) Hence write down each of the matrices $\mathbf{U}$, $\mathbf{D}$ and $\mathbf{U}^{-1}$ such that $\mathbf{A} = \mathbf{UDU}^{-1}$, where $\mathbf{D}$ is a diagonal matrix.

(4)

(b) A 2 × 2 matrix $\mathbf{M}$ has distinct real eigenvalues γ and μ, with corresponding eigenvectors $\mathbf{v}_1$ and $\mathbf{v}_2$.

(i) By considering the diagonalised form of $\mathbf{M}$, determine the eigenvalues of $\mathbf{M}^3$.

(2)

(ii) Write down the eigenvectors of $\mathbf{M}^3$

(1)
(Total 13 marks)

Q7.

The plane transformation T has matrix $\mathbf{A} = \begin{bmatrix} 1 & -2 \\ 2 & 1 \end{bmatrix}$, and maps points (x, y) onto image points (X, Y) such that

$$\begin{bmatrix} X \\ Y \end{bmatrix} = \mathbf{A} \begin{bmatrix} x \\ y \end{bmatrix}$$

(a) (i) Find A^{-1}.

(2)

(ii) Hence express each of x and y in terms of X and Y.

(2)

(b) Give a full geometrical description of T.

(5)

(c) Any plane curve with equation of the form $\frac{x^2}{p} + \frac{y^2}{q} = 1$, where p and q are distinct positive constants, is an ellipse.

(i) Show that the curve E with equation $6x^2 + y^2 = 3$ is an ellipse.

(1)

(ii) Deduce that the image of the curve E under T has equation

$$2X^2 + 4XY + 5Y^2 = 15$$

(2)

(iii) Explain why the curve with equation $2x^2 + 4xy + 5y^2 = 15$ is an ellipse.

(1)
(Total 13 marks)

Q8.

The 2 × 2 matrix **M** has an eigenvalue 3, with corresponding eigenvector $\begin{bmatrix} 1 \\ 2 \end{bmatrix}$, and a second eigenvalue –3, with corresponding eigenvector $\begin{bmatrix} 1 \\ 4 \end{bmatrix}$.

The diagonalised form of **M** is $\mathbf{M} = \mathbf{U}\,\mathbf{D}\,\mathbf{U}^{-1}$.

(a) (i) Write down suitable matrices **D** and **U**, and find $\mathbf{U}^{-1}$.

(4)

(ii) Hence determine the matrix **M**.

(3)

(b) Given that n is a positive integer, use the result $\mathbf{M}^n = \mathbf{U}\,\mathbf{D}^n\,\mathbf{U}^{-1}$ to show that:

(i) when n is even, $\mathbf{M}^n = 3^n\,\mathbf{I}$;

(ii) when n is odd, $\mathbf{M}^n = 3^{n-1}\,\mathbf{M}$.

(6)
(Total 13 marks)

Q9.

A plane transformation is represented by the 2 × 2 matrix **M**. The eigenvalues of **M** are 1 and 2,

with corresponding eigenvectors $\begin{bmatrix}1\\0\end{bmatrix}$ and $\begin{bmatrix}1\\1\end{bmatrix}$ respectively.

(a) State the equations of the invariant lines of the transformation and explain which of these is also a line of invariant points.

(3)

(b) The diagonalised form of **M** is $\mathbf{M} = \mathbf{U}\,\mathbf{D}\,\mathbf{U}^{-1}$, where **D** is a diagonal matrix.

(i) Write down a suitable matrix **D** and the corresponding matrix **U**.

(2)

(ii) Hence determine **M**.

(4)

(iii) Show that $\mathbf{M}^n = \begin{bmatrix}1 & f(n)-1\\0 & f(n)\end{bmatrix}$ for all positive integers n, where f(n) is a function of n to be determined.

(3)
(Total 12 marks)

Q10.

(a) Find the eigenvalues and corresponding eigenvectors of the matrix

$$\mathbf{X} = \begin{bmatrix}1 & 2\\5 & 4\end{bmatrix}$$

(6)

(b) (i) Write down a diagonal matrix **D**, and a suitable matrix **U**, such that

$\mathbf{X} = \mathbf{U}\mathbf{D}\mathbf{U}^{-1}$

(2)

(ii) Write down also the matrix $\mathbf{U}^{-1}$.

(1)

(iii) Use your results from parts (b)(i) and (b)(ii) to determine the matrix $\mathbf{X}^5$ in the form $\begin{bmatrix} a & b \\ c & d \end{bmatrix}$, where a, b, c and d are integers.

(3)

(Total 12 marks)

Q11.

The matrix **M** is given by $\begin{bmatrix} 2 & 7 \\ 4 & k \end{bmatrix}$, where k is a constant. It is given that the vector $\begin{bmatrix} 1 \\ 1 \end{bmatrix}$ is an eigenvector of **M**.

(a) Show that $k = 5$ and find the eigenvalue corresponding to the eigenvector $\begin{bmatrix} 1 \\ 1 \end{bmatrix}$.

(3)

(b) Determine the second eigenvalue of **M** and find a corresponding eigenvector.

(5)

(c) Write down matrices **U** and **D**, having integer elements, such that **M** can be expressed in the diagonalised form $\mathbf{M} = \mathbf{UDU}^{-1}$.

(2)

(d) Write down the matrix $\mathbf{U}^{-1}$.

(1)

(e) The matrix $\mathbf{M}^{2n}$, for positive integers n, is such that

$$\mathbf{M}^{2n} = \begin{bmatrix} a & b \\ c & d \end{bmatrix}$$

for integers a, b, c and d. Show that

$$a = p \times 4^n + q \times 81^n$$

where p and q are rational numbers to be determined.

(4)

(Total 15 marks)